

Component C - Week 1 Hand Calculations

1. Winner-Correct Rate

The true AUC values are:

Classifier	True AUC
c1	0.85
c2	0.82
c3	0.76

Therefore, the true winner is c1.

Winner-correct rate = Number of correct winners / Number of valid replicates

n_sub = 100

There are 20 valid replicates.

Replicate 1

c1

$$\begin{aligned} & (0.81 + 0.78 + 0.84 + 0.80 + 0.82) / 5 \\ & = 4.05 / 5 \\ & = 0.8100 \end{aligned}$$

c2

$$\begin{aligned} & (0.86 + 0.83 + 0.80 + 0.85 + 0.81) / 5 \\ & = 4.15 / 5 \\ & = 0.8300 \end{aligned}$$

c3

$$\begin{aligned} & (0.74 + 0.77 + 0.72 + 0.79 + 0.75) / 5 \\ & = 3.77 / 5 \\ & = 0.7540 \end{aligned}$$

Observed winner = c2

True winner = c1

Winner correct = 0

Results across 20 replicates:

Replicate	Observed Winner	Correct
1	c2	0
2	c1	1
3	c1	1
4	c1	1
5	c1	1
6	c1	1
7	c1	1

Replicate	Observed Winner	Correct
8	c1	1
9	c1	1
10	c1	1
11	c1	1
12	c2	0
13	c1	1
14	c1	1
15	c1	1
16	c1	1
17	c1	1
18	c1	1
19	c1	1
20	c1	1

Correct winners = 18

Incorrect winners = 2

Total valid replicates = 20

Winner-correct rate = $18 / 20 = 0.90$

Winner-correct rate = 0.90

n_sub = 500

There are 20 valid replicates.

Replicate 1

c1

$$(0.8443 + 0.8532 + 0.8506 + 0.8465 + 0.8488) / 5$$

$$= 4.2434 / 5$$

$$= 0.84868$$

$$= 0.8487$$

c2

$$(0.8170 + 0.8218 + 0.8377 + 0.7892 + 0.8172) / 5$$

$$= 4.0829 / 5$$

$$= 0.81658$$

$$= 0.8166$$

c3

$$(0.7621 + 0.7636 + 0.7555 + 0.7389 + 0.7639) / 5$$

$$= 3.7840 / 5$$

$$= 0.7568$$

Observed winner = c1

True winner = c1

Winner correct = 1

All 20 replicates have c1 as the observed winner.

Correct winners = 20

Incorrect winners = 0

Total valid replicates = 20

Winner-correct rate = $20 / 20 = 1.00$

Winner-correct rate = 1.00

2. Kendall's Tau

The calculation below uses replicate 1 at $n_{\text{sub}} = 100$.

True ranking: $c1 > c2 > c3$

Observed ranking: $c2 > c1 > c3$

Pair	True Order	Observed Order	Result
c1, c2	$c1 > c2$	$c2 > c1$	Discordant
c1, c3	$c1 > c3$	$c1 > c3$	Concordant
c2, c3	$c2 > c3$	$c2 > c3$	Concordant

Concordant pairs, $C = 2$

Discordant pairs, $D = 1$

Kendall's tau = $(C - D) / (C + D)$

= $(2 - 1) / (2 + 1)$

= $1 / 3 = 0.3333$

Kendall's tau = 0.3333

3. Optimism

The calculation below uses replicate 1 at $n_{\text{sub}} = 100$.

Classifier	Mean AUC
c1	0.810
c2	0.830
c3	0.754

Declared winner = c2

Observed AUC of c2 = 0.830

True AUC of c2 = 0.820

Optimism = Observed AUC of declared winner - True AUC of the same classifier

Optimism = $0.830 - 0.820 = 0.0100$

Optimism = 0.0100

4. Power: Paired t-test for c1 vs c2

The five fold scores are paired using the same fold_id.

Difference = $\text{AUC}(c1) - \text{AUC}(c2)$

Fold	c1	c2	Difference (c1 - c2)
1	0.81	0.86	-0.05
2	0.78	0.83	-0.05
3	0.84	0.80	+0.04
4	0.80	0.85	-0.05
5	0.82	0.81	+0.01

Mean Difference

$$\bar{d} = (-0.05 - 0.05 + 0.04 - 0.05 + 0.01) / 5$$

$$\bar{d} = -0.10 / 5 = -0.02$$

Variance

Fold	Difference	Difference - Mean	Squared Deviation
1	-0.05	-0.03	0.0009
2	-0.05	-0.03	0.0009
3	+0.04	+0.06	0.0036
4	-0.05	-0.03	0.0009
5	+0.01	+0.03	0.0009

Sum of squared deviations = 0.0072

$$\text{Sample variance: } s^2 = 0.0072 / (5 - 1) = 0.0018$$

$$\text{Sample standard deviation: } s = \sqrt{0.0018} = 0.04243$$

Standard Error

$$SE = s / \sqrt{5} = 0.04243 / \sqrt{5} = 0.01897$$

t-statistic

$$t = \bar{d} / SE = -0.02 / 0.01897 = -1.054$$

Degrees of freedom: $df = 5 - 1 = 4$

For a two-tailed test at $\alpha = 0.05$ with 4 degrees of freedom, the critical t-value is 2.776.

$$|t| = 1.054 < 2.776$$

The result is not statistically significant.

Two-tailed p-value = 0.351

Since $p > 0.05$: Classification = No detection

5. Extended Results

The Python implementation was run across all 20 replicates at both sample sizes.

Statistic	n_sub = 100	n_sub = 500
Winner-correct rate	0.90	1.00
Mean Kendall's tau	0.9333	1.0000
Mean optimism	0.004081	-0.001884
Power, c1 vs c2	0.30	0.85

Statistic	n_sub = 100	n_sub = 500
Sign-error rate, c1 vs c2	0.00	0.00

The larger sample size produces more stable rankings and higher power for detecting the difference between c1 and c2.

6. Code Verification

The hand calculations for replicate 1 at n_sub = 100 were checked against the Python statistics code.

Quantity	Hand Calculation	Code Output	Match
Winner-correct	0	0	Yes
Kendall's tau	0.3333	0.3333	Yes
Optimism	0.0100	0.0100	Yes
Power classification, c1 vs c2	No detection	No detection	Yes

The Python code reproduces the hand-calculated values for all four statistics for replicate 1 at n_sub = 100.